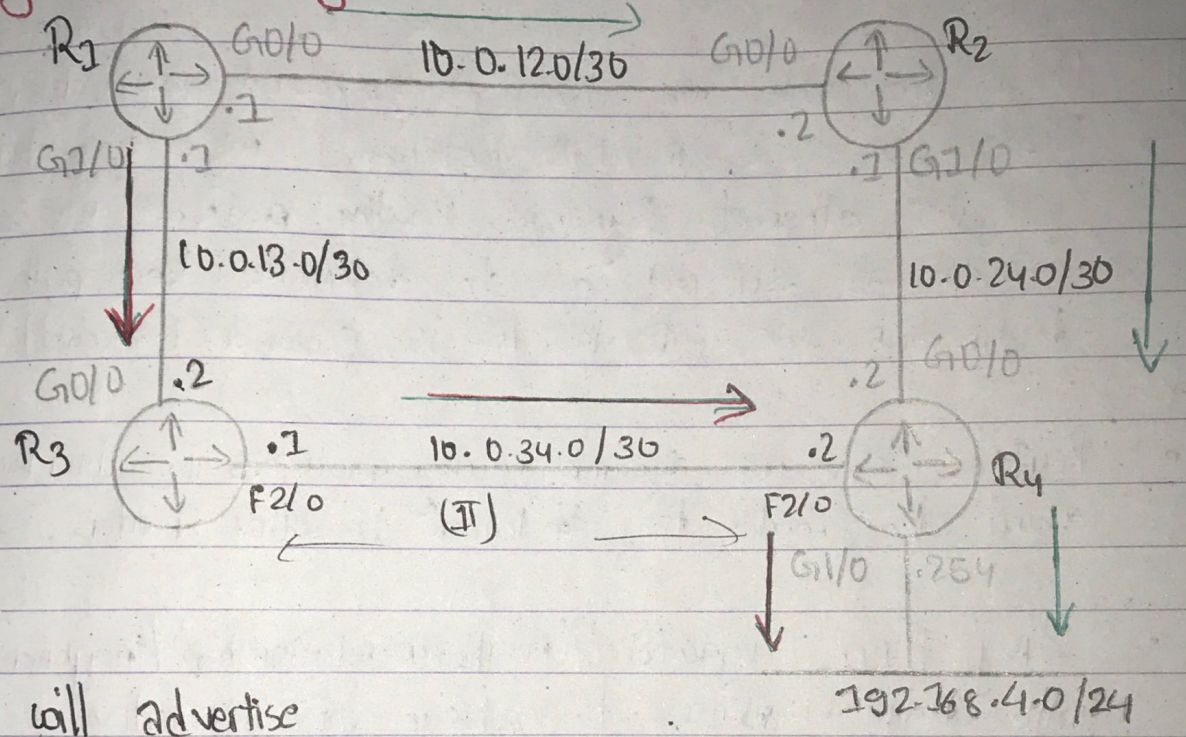
Day-24

Dynamic Routing

Network Route: A route to a network/subnet. (mask length < /32)
Host Route : A route to a specific host. (/32 mask)

Dynamic Routing

(I)



R4 will advertise its own network & LAN (192.168.40/24) to R2. which then advertise to R1 & R1 → R3, telling R3 that it can reach LAN via R1.

What if R4 G0/0 goes down? Other routers will automatically ~~add~~ adapt & remove the route from their route tables. This'll prevent other routers from continuously sending traffic to dead end.

(R4-G0/0 failure)

What if same situation happened when using static routing?

As there is no dynamic routing, R1 would be unaware of failure. So, it will continue forwarding traffic destined for LAN do R2.

Thats a benefit of dynamic routing. The router'll remove invalid routes.

Now as we add link between R3 & R4 and check ip routes between of R1 it still shows ~~a~~ p from ar→ R2→R4
And if R1 G0/0 interface is down, the ~~interface will~~ the route will shift to R1→R3→R4

Why the route via R_2 was preferred over route via R3?
↳ Thats because $R_3 \rightarrow R_4$ has Fastethernet while $R_2 - R_4$ has gigabit ethernet. Dynamic Routing protocols use similar concept as STP root cost to determine best path to destination. The path via R2 is superior (low cost)

NOB:

- Routers can use dynamic routing protocols to advertise information about the routes they ~~to~~ know to other routers.
- They form 'adjacencies'/'neighbour relationships'/'neighborships' with adjacent routers to exchange this information.
- If multiple routes to a destination are learned, the router determines which route is superior & adds it to the routing table. It uses 'metric' of the route to decide which is superior (Lower metric = superior).

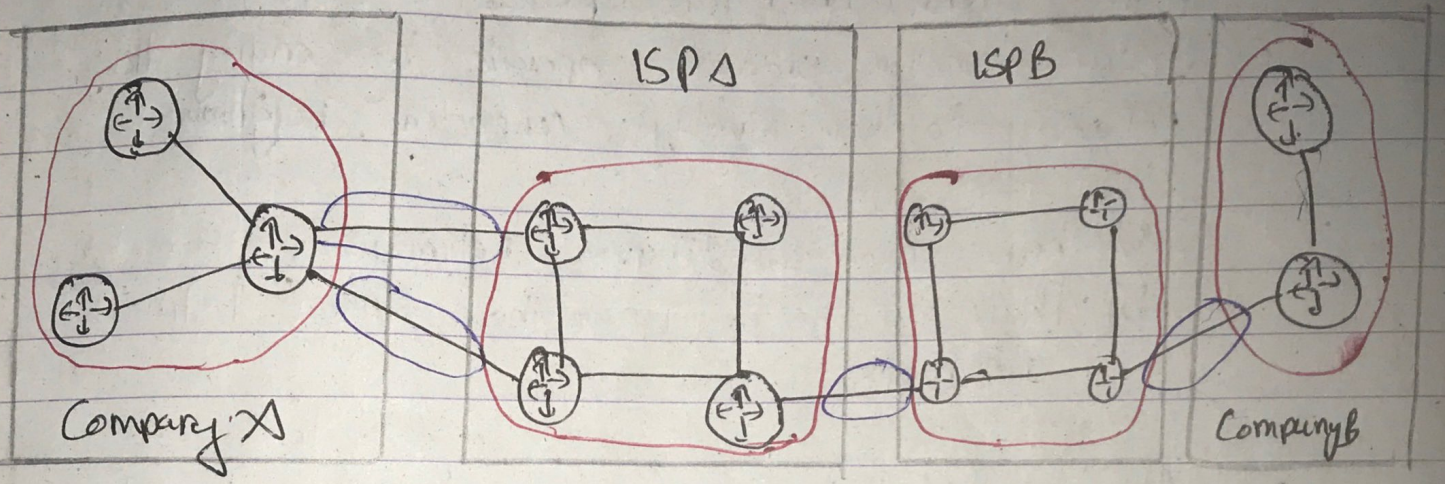
Types of Dynamic Routing Protocols.

Dynamic routing protocols can be divided into two main categories:
IGP (Interior Gateway Protocol)
EGP (Exterior Gateway Protocol)

IGP.

IGP = used to share routes within a single autonomous system (AS), which is a single organization (i-e a company)

EGP = used to share routes between different autonomous systems.



IGP — Distance Vector — Routing Information Protocol (RIP)

Enhanced Interior Gateway Routing Protocol (EIGRP)

Link State. — Open Shortest Path First (OSPF).

Intermediate System to Intermediate System (IS-IS)

EGP — Path Vector — Border Gateway Protocol (BGP)

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Distance Vector Routing Protocols

- Distance vector protocols were invented before link state protocols.
- Early examples are RIP v1 and Cisco's proprietary protocol IGRP (which was updated to EIGRP).
- Distance vector products operate by sending the following to their directly connected neighbours

  a) their known destination networks
  b) their metric to reach their known destination networks

- This method of sharing route information is often called 'routing by rumor'.
- This is because the router doesn't know about the network beyond its neighbours. It only knows the information that its neighbour told it.
- Called 'distance vector' because the routers only learn the 'distance' (metric) and 'vector' (direction the next-hop router) of each route.

Link State Routing Protocol

- When using a link state routing protocol, every router creates a 'connectivity map' of the network.
- To allow this, each router advertises information about its interfaces (connected-networks) to its neighbors. These advertisements are passed along to other routers, until all routers in the network develop the same map of network.
- Each router independently uses this map to calculate the best routes to each destination.
- Link state protocols use more resources (CPU) on the router, because more info is shared
- However, link state protocols tend to be faster in reacting to changes in the network than distance vector protocols

Dynamic Routing Protocols Metrics

- A router's route table ~~consists~~ contains the best route to each destination network it knows about
- If a router using a dynamic routing protocol learns two different routes to same destination, how does it determine which is best?
- It uses the metric value of the routes to ~~learn~~ determine which is best. A lower metric = ~~best~~ better.

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- Each routing protocol uses a different metric to determine which route is best.

NOB:

If a router learns two (or more) routes via the same routing protocol to the same destination (same network add, same subnet mask) with the same metric, both will be added to routing table. Traffic'll be load-balanced over both routes

This is called 'ECMP (Equal Cost Multi-Path)'

This can be done with static routing too

IGP	Metric	Explanation.
RIP	Hop Count	Each router in the path counts as one 'hop'. The total metric is the total number of hops to destination. Link of all speeds are equal
EIGRP	Metric based on bandwith & delay (by default).	Complex formula that can take into account many values. By default, the bandwidth of slowest link in the route & total delay of all links in the route are used.
OSPF	Cost	The cost of each link is calculated based on bandwidth. The total metric is the total cost of each link in route

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IS-IS	Cost	The total metric is the total cost of each link in route. The cost of each link isn't automatically calculated by default. All links have a cost of 10 by default.

Administrative Distance

- In most cases company will only use a single IGP - usually OSPF or EIGRP.
- However in some rare cases they might use two. For example, if two companies connect their networks to share information two different routing protocols might be in use.

- Metric is used to compare routes learned via the same routing protocol
- Different routing protocols use totally different metrics, so they can't be compared
- For example, an OSPF route to 192.168.4.0/24 might have a metric of 30, which an EIGRP route to same destination might have a metric of 33280. Which route is better? Which route should the router put in the route table?

- The administrative distance (AD) is used to determine which routing protocol is preferred.

- A lower AD is preferred, and indicates that the routing protocol is considered more 'trustworthy' (more likely to select good routes)

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If Administrative distance is 255, the router doesn't believe the source of that route & doesn't install the route in routing table

Route Protocol / Type	AD
Directly Connected	0
Static	1
External BGP (eBGP)	20
EIGRP	90
IGRP	100
OSPF	110

Route Protocol / Type	AD
IS-IS	
RIP	
EIGRP (external)	
Internal BGP (iBGP)	
Unusable route	

Administrative Distance (Cont'd...)

- The following routes to the destination network 10.1.1.0/24 are learned:

↳ next hop 192.168.1.1 learned via RIP, metric 5.
↳ next hop 192.168.2.1 learned via RIP, metric 3
↳ next hop 192.168.3.1 learned via OSPF, metric 10.
Which route to 10.1.1.0/24, will be added to the route table

- Metric is used to compare routes learned from the same routing protocol
- However, befor comparing metrics, AD is used to select the best route
- The OSPF route will always take precedence over the RIP routes, because it has lower AD.